

Thursday Reading Assessment: Chapters 4-3 through 4-10

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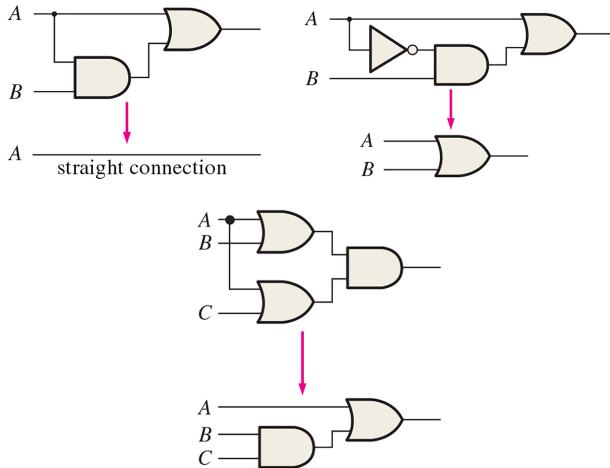


Figure 1: (Top left) Show that $A + AB = A$. (Top right) Show that $A + \bar{A}B = A + B$. (Bottom) Show that $(A + B)(A + C) = A + BC$.

1 Logic Circuits and Boolean Algebra

- Consider the three circuits in Fig. 1. Prove that each reduces to a simpler circuit using (i) a truth table, and (ii) boolean algebra.

- Consider the following equation:

$$X = \overline{\overline{A + BC} + D(E + F)} \quad (1)$$

- Use DeMorgan's Theorems to reduce this expression.
- Draw a circuit corresponding to the *unsimplified* expression.
- Draw a circuit corresponding

TABLE 4-16

Inputs			Output
A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Figure 2: A logic function definition.

to the *simplified* expression. (d) How many gates are removed from the design through simplification?

- (a) Reduce the function specified in the truth table in Fig. 2 to its minimum SOP form by using a Karnaugh map. (b) Draw the circuit.